

mathematical. The most physical book is the one by Timoshenko and Goodier. In ascending order of mathematical content are Sokolnikoff, Gurtin, Duvaut-Lions, and Fichera.

The reader is reminded that indices i, j, k , and l take on values $1, \dots, n_{sd}$, where n_{sd} is the number of spatial dimensions, and the summation convention applies to repeated indices i, j, k , and l only.

Let σ_{ij} denote (Cartesian components of) the (Cauchy) *stress tensor*, let u_i denote the *displacement vector*, and let f_i be the *prescribed body force per unit volume*. The (*infinitesimal*) *strain tensor*, ϵ_{ij} , is defined to be the symmetric part of the displacement gradients, viz.,

$$\epsilon_{ij} = u_{(i,j)} \stackrel{\text{def.}}{=} \frac{u_{i,j} + u_{j,i}}{2} \quad (\text{strain-displacement equations}) \quad (2.7.1)$$

The stress tensor is defined in terms of the strain tensor by the *generalized Hooke's law*.⁹

$$\sigma_{ij} = c_{ijkl} \epsilon_{kl} \quad (2.7.2)$$

where the c_{ijkl} 's, the *elastic coefficients*, are given functions of $\mathbf{x}$. (If the c_{ijkl} 's are constants throughout, the body is called *homogeneous*.) The elastic coefficients are assumed to satisfy the following properties:

Symmetry

$$c_{ijkl} = c_{klij} \quad (\text{major symmetry}) \quad (2.7.3)$$

$$\left. \begin{aligned} c_{ijkl} &= c_{jikl} \\ c_{ijkl} &= c_{ijlk} \end{aligned} \right\} \quad (\text{minor symmetries}) \quad (2.7.4)$$

Positive-definiteness

$$c_{ijkl}(\mathbf{x}) \psi_{ij} \psi_{kl} \geq 0 \quad (2.7.5)$$

$$c_{ijkl}(\mathbf{x}) \psi_{ij} \psi_{kl} = 0 \quad \text{implies} \quad \psi_{ij} = 0 \quad (2.7.6)$$

for all $\mathbf{x} \in \bar{\Omega}$ and all $\psi_{ij} = \psi_{ji}$.

Note. The positive-definiteness condition is in terms of symmetric arrays, ψ_{ij} .

We shall see in Sec. 2.8 that a consequence of the major symmetry (2.7.3) is that $\mathbf{K}$ is symmetric. The first minor symmetry implies the symmetry of the stress tensor (i.e., $\sigma_{ij} = \sigma_{ji}$).¹⁰ The positive-definiteness condition, when combined with appropriate boundary conditions on the displacement, leads to the positive-definiteness of $\mathbf{K}$.

⁹This is another *constitutive equation*, which reflects the elastic properties of the body under consideration.

¹⁰From a fundamental continuum mechanics standpoint, the symmetry of the Cauchy stress tensor emanates from the balance of angular momentum.

In the present theory, the unknown is a vector (i.e., the displacement vector). Consequently, a generalization of the boundary conditions considered previously is necessitated. We shall assume that Γ admits decompositions

$$\left. \begin{aligned} \Gamma &= \overline{\Gamma_{q_i} \cup \Gamma_{h_i}} \\ \emptyset &= \Gamma_{q_i} \cap \Gamma_{h_i} \end{aligned} \right\} \quad i = 1, \dots, n_{sd} \quad (2.7.7)$$

For example, in two dimensions the situation might appear as in Fig. 2.7.1. As can be seen there can be a different decomposition for each $i = 1, 2, \dots, n_{sd}$.

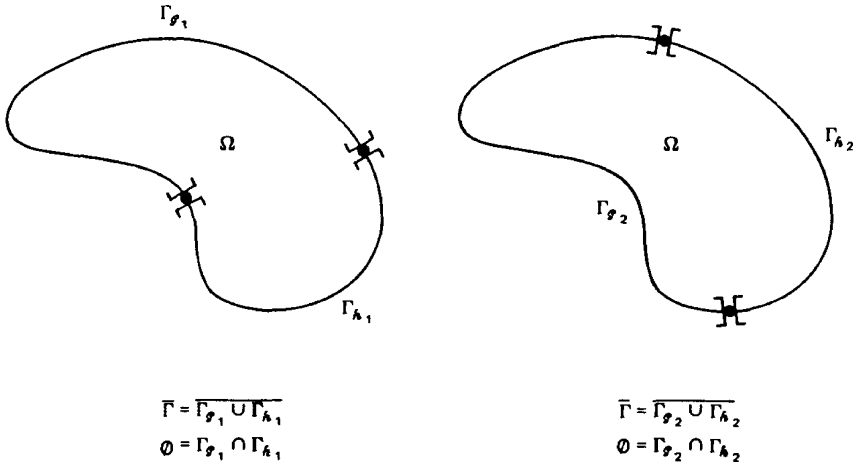


Figure 2.7.1

A formal statement of the strong form of the boundary-value problem goes as follows:

$$(S) \left\{ \begin{aligned} &\text{Given } \ell_i : \Omega \rightarrow \mathbb{R}, q_i : \Gamma_{q_i} \rightarrow \mathbb{R}, \text{ and } h_i : \Gamma_{h_i} \rightarrow \mathbb{R}, \text{ find } u_i : \bar{\Omega} \rightarrow \mathbb{R} \text{ such that} \\ &\quad \sigma_{ij,j} + \ell_i = 0 \quad \text{in } \Omega \quad (\text{equilibrium equations}) \quad (2.7.8) \\ &\quad u_i = q_i \quad \text{on } \Gamma_{q_i} \quad (2.7.9) \\ &\quad \sigma_{ij} n_j = h_i \quad \text{on } \Gamma_{h_i} \quad (2.7.10) \end{aligned} \right. \quad \text{where } \sigma_{ij} \text{ is defined in terms of } u_i \text{ by (2.7.1) and (2.7.2).}$$

Remarks

1. The functions q_i and h_i are called the *prescribed boundary displacements* and *tractions*, respectively.

2. (S) is sometimes referred to as the *mixed boundary-value problem of linear elastostatics*. Under appropriate hypotheses on the data, (S) possesses a unique solution (see Fichera [4]).

3. The additional complexity of the present theory, when compared with heat conduction, is that the unknown (i.e., $\mathbf{u} = \{u_i\}$) is vector-valued rather than scalar-valued.

4. In practice, it is important to be able to deal with somewhat more complicated boundary-condition specification. In order not to encumber the present exposition, this generalization will be considered later on in the form of an exercise (see Exercise 5 in Sec. 2.12).

Let $\mathcal{S}_i$ denote the trial solution space and $\mathcal{U}_i$ the variation space. Each member $u_i \in \mathcal{S}_i$ satisfies $u_i = q_i$ on Γ_{q_i} , whereas each $w_i \in \mathcal{U}_i$ satisfies $w_i = 0$ on Γ_{q_i} . Equation (2.7.10) will be a natural boundary condition.

The weak formulation goes as follows:

$$(W) \left\{ \begin{array}{l} \text{Given } \ell_i : \Omega \rightarrow \mathbb{R}, q_i : \Gamma_{q_i} \rightarrow \mathbb{R} \text{ and } h_i : \Gamma_{h_i} \rightarrow \mathbb{R}, \text{ find } u_i \in \mathcal{S}_i \text{ such that for all } w_i \in \mathcal{U}_i, \\ \int_{\Omega} w_{(i,j)} \sigma_{ij} d\Omega = \int_{\Omega} w_i \ell_i d\Omega + \sum_{i=1}^{n_{sd}} \left(\int_{\Gamma_{h_i}} w_i h_i d\Gamma \right) \end{array} \right. \quad (2.7.11)$$

where σ_{ij} is defined in terms of u_i by (2.7.1) and (2.7.2).

Remarks

5. In the solid mechanics literature, (W) is sometimes referred to as the *principle of virtual work*, or *principle of virtual displacements*, w_i being the *virtual displacements*.

6. The existence and uniqueness of weak solutions is discussed in Duvaut-Lions [3].

Note. The boundary integral in (2.7.11) takes on the explicit form:

$$\sum_{i=1}^{n_{sd}} \left(\int_{\Gamma_{h_i}} w_i h_i d\Gamma \right) = \int_{\Gamma_{h_1}} w_1 h_1 d\Gamma + \cdots + \int_{\Gamma_{h_{n_{sd}}}} w_{n_{sd}} h_{n_{sd}} d\Gamma \quad (2.7.12)$$

Theorem. Assume all functions are smooth enough to justify the manipulations. Then a solution of (S) is a solution of (W), and vice versa.

Remark

7. The proof of the equivalence theorem requires some preliminary results, which we shall establish in the following lemmas.

Lemma 1. (*Euclidean decomposition of a second-rank tensor.*) Let s_{ij} denote a general nonsymmetric, second-rank tensor. Then $s_{ij} = s_{(ij)} + s_{[ij]}$, where $s_{(ij)}$ is symmetric (i.e., $s_{(ij)} = s_{(ji)}$) and $s_{[ij]}$ is skew-symmetric (i.e., $s_{[ij]} = -s_{[ji]}$).

Proof. Define

$$s_{(ij)} = \frac{s_{ij} + s_{ji}}{2} \quad (2.7.13)$$

$$s_{[ij]} = \frac{s_{ij} - s_{ji}}{2} \quad (2.7.14)$$

It is easily verified that $s_{(ij)}$ is symmetric and $s_{[ij]}$ is skew-symmetric. ■

Remark

8. $s_{(ij)}$ and $s_{[ij]}$ are called the *symmetric* and *skew-symmetric parts* of s_{ij} , respectively.

Lemma 2. Let s_{ij} be a nonsymmetric tensor and let t_{ij} be a symmetric tensor. Then

$$s_{ij}t_{ij} = s_{(ij)}t_{ij} \quad (2.7.15)$$

Proof. By Lemma 1, $s_{ij} = s_{(ij)} + s_{[ij]}$. Since

$$s_{ij}t_{ij} = s_{(ij)}t_{ij} + s_{[ij]}t_{ij}$$

the lemma will be established if we can show that $s_{[ij]}t_{ij} = 0$. We proceed as follows:

$$\begin{aligned} s_{[ij]}t_{ij} &= -s_{[ji]}t_{ij} && \text{(skew-symmetry of } s_{[ij]}) \\ &= -s_{[ij]}t_{ji} && \text{(symmetry of } t_{ij}) \\ &= -s_{[ij]}t_{ij} && \text{(redefinition of dummy indices)} \end{aligned}$$

from which the result follows. ■

Proof of Theorem

1. Let u_i be a solution of (S). Thus $u_i \in \mathcal{S}_i$. Multiply (2.7.8) by $w_i \in \mathcal{U}_i$ and integrate over Ω , viz.,

$$\begin{aligned} 0 &= \int_{\Omega} w_i(\sigma_{ij,j} + \ell_i) d\Omega = - \int_{\Omega} w_{i,j} \sigma_{ij} d\Omega + \int_{\Gamma} w_i \sigma_{ij} n_j d\Gamma + \int_{\Omega} w_i \ell_i d\Omega \\ &\hspace{25em} \text{(integration by parts)} \\ &= - \int_{\Omega} w_{(i,j)} \sigma_{ij} d\Omega + \sum_{i=1}^{nd} \left(\int_{\Gamma_{q_i}} w_i h_i d\Gamma \right) + \int_{\Omega} w_i \ell_i d\Omega \quad \begin{array}{l} \text{(symmetry of } \sigma_{ij}, \\ \text{Lemma 2, } w_i = 0 \\ \text{on } \Gamma_{q_i}, \text{ and (2.7.10))} \end{array} \end{aligned}$$

Therefore, u_i is a solution of (W).

2. Assume u_i is a solution of (W). Since $u_i \in \mathcal{S}_i$, $u_i = q_i$ on Γ_{q_i} . From (2.7.11)

$$\begin{aligned} 0 &= - \int_{\Omega} \underbrace{w_{(i,j)} \sigma_{ij}}_{(= w_{i,j} \sigma_{ij} \text{ by Lemma 2})} d\Omega + \int_{\Omega} w_i \ell_i d\Omega + \sum_{i=1}^{nd} \int_{\Gamma_{q_i}} w_i h_i d\Gamma \end{aligned}$$

$$= \int_{\Omega} w_i (\sigma_{ij,j} + \ell_i) d\Omega - \sum_{i=1}^{n_M} \int_{\Gamma_{k_i}} w_i (\sigma_{ij} n_j - h_i) d\Gamma \quad \begin{array}{l} \text{(integration by parts,} \\ w_i = 0 \text{ on } \Gamma_{q_i}) \end{array} \quad (2.7.16)$$

Let

$$\alpha_i = \sigma_{ij,j} + \ell_i$$

$$\beta_i = \sigma_{ij} n_j - h_i$$

Thus to complete the proof we must show that

$$\alpha_i = 0 \quad \text{on } \Omega$$

$$\beta_i = 0 \quad \text{on } \Gamma_{k_i}$$

These conditions can be proved by the techniques used in Sec. 2.3. Let $w_i = \alpha_i \phi$, where

- i. $\phi > 0$ on Ω ;
- ii. $\phi = 0$ on Γ ; and
- iii. ϕ is smooth.

(These conditions insure that $w_i \in \mathcal{U}_i$.) Substituting this w_i into (2.7.16) yields

$$0 = \int_{\Omega} \underbrace{\alpha_i \alpha_i}_{\geq 0} \underbrace{\phi}_{> 0} d\Omega$$

which implies $\alpha_i = 0$ on Ω .

Now take $w_i = \delta_{i1} \beta_1 \psi$, where

- i'. $\psi > 0$ on Γ_{k_1} ;
- ii'. $\psi = 0$ on Γ_{q_1} ; and
- iii'. ψ is smooth.

(Again, $w_i \in \mathcal{U}_i$.) Substituting this w_i into (2.7.16) and making use of $\alpha_i = 0$ results in

$$0 = \int_{\Gamma_{k_1}} \beta_1^2 \psi d\Gamma$$

from which it follows that $\beta_1 = 0$ on Γ_{k_1} .

We may proceed analogously to show $\beta_2 = 0$, and so on. Consequently, u_i is a solution of $(\bar{S})$. ■

The abstract notation for the present case is

$$a(w, u) = \int_{\Omega} w_{(i,j)} c_{ijkl} u_{(k,l)} d\Omega \quad (2.7.17)$$

$$(w, \ell) = \int_{\Omega} w_i \ell_i d\Omega \quad (2.7.18)$$

$$(w, h)_{\Gamma} = \sum_{i=1}^{n_{sd}} \left(\int_{\Gamma_{k_i}} w_i h_i d\Gamma \right) \quad (2.7.19)$$

Exercise 1. Verify that $a(\cdot, \cdot)$, $(\cdot, \cdot)$ and $(\cdot, \cdot)_{\Gamma}$, as just defined, are symmetric, bilinear forms (cf. Sec. 1.4).

Let $\mathcal{S} = \{u \mid u_i \in \mathcal{S}_i\}$ and let $\mathcal{V} = \{w \mid w_i \in \mathcal{V}_i\}$. Then the weak form can be concisely written in terms of (2.7.17–2.7.19) as follows:

$$(W) \left\{ \begin{array}{l} \text{Given } \ell, q, \text{ and } h \text{ (in which the components are defined in } (W)), \text{ find } u \in \mathcal{S} \\ \text{such that for all } w \in \mathcal{V} \end{array} \right. \quad \boxed{a(w, u) = (w, \ell) + (w, h)_{\Gamma}} \quad (2.7.20)$$

As discussed in Sec. 2.3, it is desirable to construct index-free counterparts of the expressions on the right-hand sides of (2.7.17)–(2.7.19). For concreteness we shall assume that $n_{sd} = 2$; thus $1 \leq i, j, k, l \leq 2$.

Let¹¹

$$\epsilon(u) = \{\epsilon_i(u)\} = \left\{ \begin{array}{c} u_{1,1} \\ u_{2,2} \\ u_{1,2} + u_{2,1} \end{array} \right\} \quad (2.7.21)$$

$$\epsilon(w) = \{\epsilon_i(w)\} = \left\{ \begin{array}{c} w_{1,1} \\ w_{2,2} \\ w_{1,2} + w_{2,1} \end{array} \right\} \quad (2.7.22)$$

$$D = [D_{IJ}] = \begin{bmatrix} D_{11} & D_{12} & D_{13} \\ & D_{22} & D_{23} \\ \text{symmetric} & & D_{33} \end{bmatrix} \quad (2.7.23)$$

¹¹ According to our previous notational conventions, $\epsilon = [\epsilon_{ij}]$, the matrix of strain components. However, we will have no need for this matrix, and consequently we reserve ϵ for the “strain vector” defined in (2.7.21). A similar notational conflict occurs with respect to the “stress vector,” σ , defined in Exercise 4. Note that factors of one-half have been eliminated from the *shearing components* (i.e., last components) of (2.7.21) and (2.7.22). (Compare (2.7.21) and (2.7.22) with (2.7.1).) This will considerably simplify subsequent writing.

where

$$D_{IJ} = c_{ijkl} \quad (2.7.24)$$

in which the indices are related by the following table:

TABLE 2.7.1

$I \backslash J$	$i \backslash k$	$j \backslash l$
1	1	1
3	1	2
3	2	1
2	2	2

As should be clear, we have “collapsed” pairs of indices (i, j , and k, l) into single indices (I and J , respectively) taking account of the symmetries of c_{ijkl} , $u_{(k,l)}$ and $w_{(i,j)}$. Observe that the indices I and J take on values 1, 2, and 3. In n_{sd} dimensions, the I and J indices will take on values 1, 2, . . . , $n_{sd}(n_{sd} + 1)/2$.

It can be shown that

$$w_{(i,j)} c_{ijkl} u_{(k,l)} = \epsilon(w)^T D \epsilon(u) \quad (2.7.25)$$

and so

$$a(w, u) = \int_{\Omega} \epsilon(w)^T D \epsilon(u) d\Omega \quad (2.7.26)$$

Exercise 2. Verify (2.7.25) for $n_{sd} = 2$.

Exercise 3. Construct the analog of Table 2.7.1 for the case $n_{sd} = 3$. For definiteness of the ordering, take

$$\epsilon(u) = \begin{Bmatrix} u_{1,1} \\ u_{2,2} \\ u_{3,3} \\ u_{2,3} + u_{3,2} \\ u_{1,3} + u_{3,1} \\ u_{1,2} + u_{2,1} \end{Bmatrix} \quad (2.7.27)$$

Exercise 4. Show that

$$\sigma = D \epsilon(u) \quad (2.7.28)$$

where

$$\boldsymbol{\sigma} = \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{Bmatrix}, \quad n_{sd} = 2 \quad (2.7.29)$$

$$\boldsymbol{\sigma} = \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{Bmatrix}, \quad n_{sd} = 3 \quad (2.7.30)$$

Exercise 5. If the body in question is *isotropic*, then

$$c_{ijkl}(x) = \mu(x)(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + \lambda(x)\delta_{ij}\delta_{kl} \quad (2.7.31)$$

where λ and μ are the *Lamé parameters*; μ is often referred to as the shear modulus and denoted by G . The relationships of λ and μ to E , *Young's modulus*, and ν , *Poisson's ratio*, are given by

$$\lambda = \frac{\nu E}{(1 + \nu)(1 - 2\nu)} \quad (2.7.32)$$

$$\mu = \frac{E}{2(1 + \nu)} \quad (2.7.33)$$

(See Sokolnikoff [6], p. 71, for further relations with other equivalent moduli.) If (2.7.31) is not satisfied, the body is said to be *anisotropic*. Using (2.7.31) set up D for $n_{sd} = 2$ and $n_{sd} = 3$. *Hint:* The answer for $n_{sd} = 2$ is

$$D = \begin{bmatrix} \lambda + 2\mu & \lambda & 0 \\ & \lambda + 2\mu & 0 \\ \text{Symmetric} & & \mu \end{bmatrix} \quad (2.7.34)$$

This matrix manifests the *plane strain* hypothesis. (See Sokolnikoff [6] for elaboration.)

Remark

9. The case of isotropic *plane stress* may be determined from (2.7.34) by replacing λ by $\bar{\lambda}$, where

$$\bar{\lambda} = \frac{2\lambda\mu}{\lambda + 2\mu} \quad (2.7.35)$$

(See Sokolnikoff [6] or Timoshenko and Goodier [7] for elaboration on the physical ideas.)

Exercise 6. (See Exercise 1, Sec. 2.4 for background and an analogous result.)

Let the "jump" in $\sigma_{in} = \sigma_{ij}n_j$ be denoted by $[\sigma_{in}]$. Consider the weak formulation (i.e., (2.7.11)) and assume w_i and u_i are smooth on element interiors, but experience gradient discontinuities across element boundaries. Show that

$$0 = \sum_{e=1}^{n_{el}} \int_{\Omega^e} w_i (\sigma_{ij,j} + f_i) d\Omega - \sum_{i=1}^{n_{sd}} \int_{\Gamma_{k_i}} w_i (\sigma_{in} - h_i) d\Gamma - \int_{\Gamma_{int}} w_i [\sigma_{in}] d\Gamma$$

from which the Euler-Lagrange conditions may be read:

- i. $\sigma_{ij,j} + f_i = 0$ in $\bigcup_{e=1}^{n_{el}} \Omega^e$
- ii. $\sigma_{in} = h_i$ on Γ_{k_i}
- iii. $[\sigma_{in}] = 0$ on Γ_{int}

Here (i) is the equilibrium equation on *element interiors*, and (iii) is a traction continuity condition across element boundaries. Compare these results with those obtained assuming w_i and u_i are *globally* smooth.

2.8 ELASTOSTATICS: GALERKIN FORMULATION, SYMMETRY, AND POSITIVE-DEFINITENESS OF K

Let $\mathcal{S}^h$ and $\mathcal{O}^h$ be finite-dimensional approximations to $\mathcal{S}$ and $\mathcal{O}$, respectively. We assume members $w^h \in \mathcal{O}^h$ result in satisfaction, or approximate satisfaction, of the boundary condition $w_i = 0$ on Γ_{q_i} , and members of $\mathcal{S}^h$ admit the decomposition

$$u^h = v^h + q^h \quad (2.8.1)$$

where $v^h \in \mathcal{O}^h$ and q^h results in satisfaction, or approximate satisfaction, of the boundary condition $u_i = q_i$ on Γ_{q_i} .

The Galerkin formulation of our problem is given as follows:

$$(G) \left\{ \begin{array}{l} \text{Given } \ell, q, \text{ and } h \text{ (as in } (W)), \text{ find } u^h = v^h + q^h \in \mathcal{S}^h \text{ such that for all } w^h \in \mathcal{O}^h \\ \quad \quad \quad a(w^h, v^h) = (w^h, \ell) + (w^h, h)_\Gamma - a(w^h, q^h) \end{array} \right. \quad (2.8.2)$$

To define the global stiffness matrix and force vector for elasticity, it is necessary to introduce the ID array. This entails a generalization of the definition given in Sec. 2.6, since in the present case there will be more than 1 degree of freedom per node. For elasticity there are n_{sd} degrees of freedom per node, but in order to include in our

definition cases such as heat conduction, we shall take the fully general situation in which it is assumed that there are n_{dof} *degrees of freedom per node*.¹² In this case

$$\text{ID}(i, A) = \begin{cases} P & \text{if } A \in \eta - \eta_{q_i} \\ 0 & \text{if } A \in \eta_{q_i} \end{cases} \quad (2.8.3)$$

Global equation number
 ↓
 Degree of freedom number Global node number

where $1 \leq i \leq n_{\text{dof}}$. Thus ID has dimensions $n_{\text{dof}} \times n_{np}$. If $n_{\text{dof}} = 1$, we reduce to the case considered previously in Sec. 2.6 (i.e., $\text{ID}(i, A) = \text{ID}(A)$).

Recall that $\eta = \{1, 2, \dots, n_{np}\}$ denotes the set of global node numbers. Let $\eta_{q_i} \subset \eta$ be the set of nodes at which $u_i^h = q_i$ and let $\eta - \eta_{q_i}$ be the complement of η_{q_i} . For each node in $\eta - \eta_{q_i}$, the nodal value of u_i^h is to be determined.

The explicit representations of v_i^h and q_i^h , in terms of the shape functions and nodal values are

$$v_i^h = \sum_{A \in \eta - \eta_{q_i}} N_A d_{iA} \quad (\text{no sum on } i) \quad (2.8.4)$$

Degree of freedom number
 ↙ ↘
 Global node number

$$q_i^h = \sum_{A \in \eta_{q_i}} N_A q_{iA} \quad (\text{no sum on } i) \quad (2.8.5)$$

Let e_i denote the i th Euclidean basis vector for $\mathbb{R}^{n_{sd}}$; e_i has a 1 in slot i and zeros elsewhere. For example

$$(n_{sd} = 2) \quad e_1 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}, \quad e_2 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} \quad (2.8.6)$$

¹²In general, this is taken to mean the maximum number of degrees of freedom per node in the global model. It is possible in practice to have elements with fewer degrees of freedom per node contributing to the model.

$$(n_{sd} = 3) \quad e_1 = \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix}, \quad e_2 = \begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix}, \quad e_3 = \begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix} \quad (2.8.7)$$

The vector versions of (2.8.4) and (2.8.5) may be defined with the aid of e_i , viz.,

$$v^h = v_i^h e_i \quad (2.8.8)$$

$$q^h = q_i^h e_i \quad (2.8.9)$$

Likewise, a typical member $w^h \in \mathcal{O}^h$ has the representation

$$w^h = w_i^h e_i, \quad w_i^h = \sum_{A \in \eta - \eta_{q_i}} N_A c_{iA} \quad (\text{no sum on } i) \quad (2.8.10)$$

Substituting (2.8.4), (2.8.5), and (2.8.8)–(2.8.10) into (2.8.2) and arguing along the lines of Sec. 1.6 results in (verify!)

$$\sum_{j=1}^{n_{\text{dof}}} \left(\sum_{B \in \eta - \eta_{q_j}} a(N_A e_i, N_B e_j) d_{jB} \right) = (N_A e_i, f) + (N_A e_i, h)_\Gamma$$

$$- \sum_{j=1}^{n_{\text{dof}}} \left(\sum_{B \in \eta_{q_j}} a(N_A e_i, N_B e_j) q_{jB} \right), \quad A \in \eta - \eta_{q_i}, \quad 1 \leq i \leq n_{sd}$$

(2.8.11)¹³

This is equivalent to the matrix equation

¹³For correct interpretation of the meaning of these equations, the sum on j should be taken first. For example, in two dimensions

$$\sum_{j=1}^2 \left(\sum_{B \in \eta - \eta_{q_j}} a(N_A e_i, N_B e_j) d_{jB} \right) = \sum_{B \in \eta - \eta_{q_1}} a(N_A e_i, N_B e_1) d_{1B}$$

$$+ \sum_{B \in \eta - \eta_{q_2}} a(N_A e_i, N_B e_2) d_{2B}$$

and

$$\sum_{j=1}^2 \left(\sum_{B \in \eta_{q_j}} a(N_A e_i, N_B e_j) q_{jB} \right) = \sum_{B \in \eta_{q_1}} a(N_A e_i, N_B e_1) q_{1B}$$

$$+ \sum_{B \in \eta_{q_2}} a(N_A e_i, N_B e_2) q_{2B}$$

$$Kd = F \quad (2.8.12)$$

where

$$K = [K_{PQ}] \quad (2.8.13)$$

$$d = \{d_Q\} \quad (2.8.14)$$

$$F = \{F_P\} \quad (2.8.15)$$

$$K_{PQ} = a(N_A e_i, N_B e_j) \quad (2.8.16)$$

$$F_P = (N_A e_i, \ell) + (N_A e_i, h)_\Gamma - \sum_{j=1}^{n_{\text{dof}}} \left(\sum_{B \in \eta_{\theta_j}} a(N_A e_i, N_B e_j) q_{jB} \right) \quad (2.8.17)^{13}$$

in which

$$P = \text{ID}(i, A), \quad Q = \text{ID}(j, B) \quad (2.8.18)$$

Equation (2.8.16) may be written in more explicit form by using (2.7.26) and noting that (see (2.7.21) and (2.7.22)):

$$\epsilon(N_A e_i) = B_A e_i \quad (2.8.19)$$

where

$$(n_{sd} = 2) \quad B_A = \begin{bmatrix} N_{A,1} & 0 \\ 0 & N_{A,2} \\ N_{A,2} & N_{A,1} \end{bmatrix} \quad (2.8.20)$$

$$(n_{sd} = 3) \quad B_A = \begin{bmatrix} N_{A,1} & 0 & 0 \\ 0 & N_{A,2} & 0 \\ 0 & 0 & N_{A,3} \\ 0 & N_{A,3} & N_{A,2} \\ N_{A,3} & 0 & N_{A,1} \\ N_{A,2} & N_{A,1} & 0 \end{bmatrix} \quad (2.8.21)$$

Exercise 1. Verify (2.8.19)–(2.8.21).

With these, (2.8.16) becomes

$$K_{PQ} = e_i^T \int_{\Omega} B_A^T D B_B d\Omega e_j \quad (2.8.22)$$

and the indices are related by (2.8.18).

Equation (2.8.17) is also amenable to explication. Note that, by (2.7.18)

$$(N_A e_i, \ell) = \int_{\Omega} N_A \ell_i d\Omega \quad (2.8.23)$$

and likewise by (2.7.19)

$$(N_A e_i, h)_{\Gamma} = \int_{\Gamma_{k_i}} N_A h_i d\Gamma \quad (\text{no sum}) \quad (2.8.24)$$

Thus (2.8.17) may be written as

$$F_P = \int_{\Omega} N_A \ell_i d\Omega + \int_{\Gamma_{k_i}} N_A h_i d\Gamma - \sum_{j=1}^{n_{\text{dof}}} \left(\sum_{B \in \eta_{\theta_j}} a(N_A e_i, N_B e_j) q_{jB} \right) \quad (2.8.25)^{14}$$

Now that we have defined $\mathbf{K}$, we can establish its fundamental properties. We shall need the following preliminary results.

Let $n_{sd} = 2$ or 3 and let $\mathbf{w} : \Omega \rightarrow \mathbb{R}^{n_{sd}}$. If $w_{(i,j)} = 0$ ("zero strains"), then $\mathbf{w}$ admits the representations:

$$(n_{sd} = 2) \quad \mathbf{w}(\mathbf{x}) = \overbrace{\mathbf{c}}^{\text{Translation}} + \overbrace{c_3(x_1 \mathbf{e}_2 - x_2 \mathbf{e}_1)}^{\text{Rotation}} \quad (2.8.26)$$

$$(n_{sd} = 3) \quad \mathbf{w}(\mathbf{x}) = \overbrace{\mathbf{c}_1}^{\text{Translation}} + \overbrace{\mathbf{c}_2 \times \mathbf{x}}^{\text{Rotation}} \quad (2.8.27)$$

where

$$\mathbf{c} = \begin{Bmatrix} c_1 \\ c_2 \end{Bmatrix} \quad \mathbf{c}_1 = \begin{Bmatrix} c_{11} \\ c_{12} \\ c_{13} \end{Bmatrix} \quad \mathbf{c}_2 = \begin{Bmatrix} c_{21} \\ c_{22} \\ c_{23} \end{Bmatrix} \quad (2.8.28)$$

and c_3 are constants; and $\times$ denotes the vector cross product. Equations (2.8.26) and (2.8.27) define *infinitesimal rigid-body motions*.

Assumption R

We assume that the homogeneous boundary conditions incorporated into the definition of $\mathcal{O}^h$ preclude nontrivial infinitesimal rigid-body motions. In other words, we assume that if $\mathbf{w}^h \in \mathcal{O}^h$ is a rigid-body motion, then $\mathbf{w}^h$ is identically zero.

Theorem

1. $\mathbf{K}$ is symmetric.
2. If Assumption R holds, then $\mathbf{K}$ is also positive definite.

Proof of 1. Symmetry We may note that symmetry of $\mathbf{K}$ follows from (2.8.16) and the symmetry of $a(\cdot, \cdot)$. However, we shall provide an alternative proof in terms of (2.8.22).

¹⁴See footnote 13 of this chapter.

$$\begin{aligned}
 K_{PQ} &= \mathbf{e}_i^T \int_{\Omega} \mathbf{B}_A^T D \mathbf{B}_B d\Omega \mathbf{e}_j \\
 &= \mathbf{e}_j^T \int_{\Omega} \mathbf{B}_B^T D^T \mathbf{B}_A d\Omega \mathbf{e}_i \\
 &= \mathbf{e}_j^T \int_{\Omega} \mathbf{B}_B^T D \mathbf{B}_A d\Omega \mathbf{e}_i \quad (\text{symmetry of } D) \\
 &= K_{QP}
 \end{aligned}$$

Remark

Note that the symmetry of $\mathbf{K}$ followed from the symmetry of D , which was a consequence of the major symmetry of the c_{ijkl} 's (see (2.7.3)).

Proof of 2. Positive definite (Recall from Sec. 1.9 that we must show (i) $\mathbf{c}^T \mathbf{K} \mathbf{c} \geq 0$ and (ii) $\mathbf{c}^T \mathbf{K} \mathbf{c} = 0$ implies $\mathbf{c} = \mathbf{0}$.)

Let $\mathbf{w}_i^h = \sum_{A \in \eta - \eta_{q_i}} N_A c_{iA}$ be a member of $\mathcal{U}_i^h$. Then $c_P = c_{iA}$, where $P = \text{ID}(i, A)$, $1 \leq P \leq n_{eq}$, defines the components of an n_{eq} -vector $\mathbf{c}$.

$$\begin{aligned}
 \mathbf{c}^T \mathbf{K} \mathbf{c} &= \sum_{P, Q=1}^{n_{eq}} c_P K_{PQ} c_Q \\
 &= \sum_{i, j=1}^{n_{dof}} \left(\sum_{\substack{A \in \eta - \eta_{q_i} \\ B \in \eta - \eta_{q_j}}} c_{iA} a(N_A \mathbf{e}_i, N_B \mathbf{e}_j) c_{jB} \right) \quad (\text{definition of } K_{PQ}) \\
 &= a \left(\sum_{i=1}^{n_{dof}} \left(\sum_{A \in \eta - \eta_{q_i}} c_{iA} N_A \mathbf{e}_i \right), \sum_{j=1}^{n_{dof}} \left(\sum_{B \in \eta - \eta_{q_j}} c_{jB} N_B \mathbf{e}_j \right) \right) \quad (\text{bilinearity of } a(\cdot, \cdot)) \\
 &= a(\mathbf{w}^h, \mathbf{w}^h) \quad (\text{definition of } \mathbf{w}^h) \\
 &= \int_{\Omega} \underbrace{\mathbf{w}_{(i,j)}^h c_{ijkl} \mathbf{w}_{(k,l)}^h}_{\geq 0} d\Omega \quad (\text{by (2.7.5) and (2.7.17)}) \\
 &\geq 0
 \end{aligned}$$

ii. Assume $\mathbf{c}^T \mathbf{K} \mathbf{c} = 0$. By the proof of part (i), we deduce that

$$\mathbf{w}_{(i,j)}^h c_{ijkl} \mathbf{w}_{(k,l)}^h = 0$$

From (2.7.6), this means that $\mathbf{w}_{(i,j)}^h = \mathbf{0}$, and so $\mathbf{w}^h$ is an infinitesimal rigid motion. By Assumption R, $\mathbf{w}^h = \mathbf{0}$, from which it follows that $c_P = 0$; hence $\mathbf{c} = \mathbf{0}$. ■

Remark

Positive definiteness of $\mathbf{K}$ is based upon two requirements: a positive-definiteness condition on the constitutive coefficients and suitable boundary conditions being incorporated into $\mathcal{U}^h$.

2.9 ELASTOSTATICS: ELEMENT STIFFNESS MATRIX AND FORCE VECTOR

As usual, K and F may be decomposed into sums of elemental contributions. These results will be omitted here as the reader should now be familiar with the ideas involved (cf. Sec. 2.5). We will proceed directly to the definitions of k^e and f^e :

$$k^e = [k_{pq}^e], \quad f^e = \{f_p^e\}, \quad 1 \leq p, q \leq n_{ee} = n_{ed}n_{en} \quad (2.9.1)^{15}$$

$$k_{pq}^e = e_i^T \int_{\Omega^e} B_a^T DB_b d\Omega e_j, \quad p = n_{ed}(a-1) + i, \quad (2.9.2)$$

$$q = n_{ed}(b-1) + j$$

$$(n_{sd} = 2) \quad B_a = \begin{bmatrix} N_{a,1} & 0 \\ 0 & N_{a,2} \\ N_{a,2} & N_{a,1} \end{bmatrix} \quad (2.9.3)$$

$$(n_{sd} = 3) \quad B_a = \begin{bmatrix} N_{a,1} & 0 & 0 \\ 0 & N_{a,2} & 0 \\ 0 & 0 & N_{a,3} \\ 0 & N_{a,3} & N_{a,2} \\ N_{a,3} & 0 & N_{a,1} \\ N_{a,2} & N_{a,1} & 0 \end{bmatrix} \quad (2.9.4)$$

and

$$f_p^e = \int_{\Omega^e} N_a t_i d\Omega + \int_{\Gamma_{k_i}^e} N_a h_i d\Gamma - \sum_{q=1}^{n_{ee}} k_{pq} q_q^e, \quad \Gamma_{k_i}^e = \Gamma_{k_i} \cap \Gamma^e \quad (2.9.5)$$

(no sum on i)

where $q_q^e = q_{jb}^e = q_j(x_b)$ if q_j is prescribed at node b , and equals zero otherwise.

It is useful for programming purposes to define the *nodal submatrix*

$$\underbrace{k_{ab}^e}_{n_{ed} \times n_{ed}} = \int_{\Omega^e} B_a^T DB_b d\Omega \quad (2.9.6a)$$

From (2.9.2) we see that

¹⁵ n_{ee} stands for the *number of element equations* and n_{ed} is the *number of element degrees of freedom* (per node). It is possible in practice to have $n_{ed} < n_{dof}$, although they are usually equal.

$$k_{pq}^e = e_i^T k_{ab}^e e_j \quad (2.9.6)$$

This means, "the pq -component of k^e is the ij -component of the submatrix k_{ab}^e ."

By (2.9.1) through (2.9.4), we see that k^e may be written as

$$k^e = \int_{\Omega^e} B^T DB \, d\Omega \quad (2.9.7)$$

where

$$B = [B_1, B_2, \dots, B_{n_{en}}] \quad (2.9.8)$$

For example, in the case of a two-dimensional (i.e., $n_{sd}=n_{ed}=2$), four-noded element k^e takes the form

$$\underbrace{k^e}_{8 \times 8} = \begin{bmatrix} k_{11}^e & k_{12}^e & k_{13}^e & k_{14}^e \\ \vdots & \vdots & \vdots & \vdots \\ k_{21}^e & k_{22}^e & k_{23}^e & k_{24}^e \\ \vdots & \vdots & \vdots & \vdots \\ k_{31}^e & k_{32}^e & k_{33}^e & k_{34}^e \\ \vdots & \vdots & \vdots & \vdots \\ k_{41}^e & k_{42}^e & k_{43}^e & k_{44}^e \end{bmatrix} \quad (2.9.9)$$

In practice, the submatrices above the dashed line are computed and those below, required, are determined by symmetry.

The global arrays K and F may be formed from the element arrays k^e and f^e respectively, by way of an assembly algorithm as outlined in Sec. 1.14.

Exercise 1. Let

$$\underbrace{d^e}_{n_{ee} \times 1} = \{d_a^e\} = \begin{Bmatrix} d_1^e \\ d_2^e \\ \vdots \\ d_{n_{en}}^e \end{Bmatrix} \quad (2.9.10)$$

$$(n_{ed} = 2) \quad d_a^e = \begin{Bmatrix} d_{1a}^e \\ d_{2a}^e \end{Bmatrix} \quad (2.9.11)$$

$$(n_{ed} = 3) \quad d_a^e = \begin{Bmatrix} d_{1a}^e \\ d_{2a}^e \\ d_{3a}^e \end{Bmatrix} \quad (2.9.12)$$

where

$$d_{ia}^e = u_i^h(x_a^e) \quad (2.9.13)^{16}$$

d^e is called the *element displacement vector*. Show that the stress vector (see Exercise 4, Sec. 2.7.) at point $x \in \Omega^e$ can be calculated from the formula

$$\sigma(x) = D(x)B(x)d^e = D(x) \sum_{a=1}^{n_{en}} B_a(x)d_a^e \quad (2.9.14)$$

2.10 ELASTOSTATICS: DATA PROCESSING ARRAYS ID, IEN, AND LM

We have already noted that the definition of the ID array must be generalized for the present case as indicated in Sec. 2.8. We must also generalize our definition of the LM array. However, the IEN array remains the same as before.

In the present and fully general cases, the LM array is three-dimensional, with dimensions $n_{ed} \times n_{en} \times n_{el}$, and is defined by

$$LM(i, a, e) = ID(i, IEN(a, e))$$

Degrees of
freedom
number

Local
node
number

Element
number

(2.10.1)

Alternatively, it is sometimes convenient to think of LM as two-dimensional, with dimensions $n_{ee} \times n_{el}$, viz.,¹⁷

$$LM(p, e) = LM(i, a, e), \quad p = n_{ed}(a - 1) + i$$

Local
equation
number

Element
number

(2.10.2)

To see how everything works in practice, it is helpful to run through a simple example.

Example 1

Consider the mesh of four-node, rectangular elements illustrated in Fig. 2.10.1. We assume that the local node numbering begins in the lower left-hand corner for each element and proceeds in counterclockwise fashion.

¹⁶ The q -boundary conditions are accounted for in this definition.

¹⁷ The reader knowledgeable in FORTRAN will realize that the internal computer storage of (2.10.1) and (2.10.2) is identical.

This is shown for element 4, which is typical. Four displacement (i.e., “ q -type” boundary conditions are specified; namely, the horizontal displacement is specified at nodes 1 and 3, and the vertical displacement is specified at nodes 1 and 3. Since $n_{np} = 12$, $n_{dof} = n_{ed} = 2$, and 4 displacement degrees of freedom are specified, we have $n_{eq} = 20$. As is usual, we adopt the convention that the global equation numbers run in ascending order with respect to the ascending order of global node numbers.¹⁸ The IEN, and LM arrays are given in Figure 2.10.2. The reader is urged to verify the results.

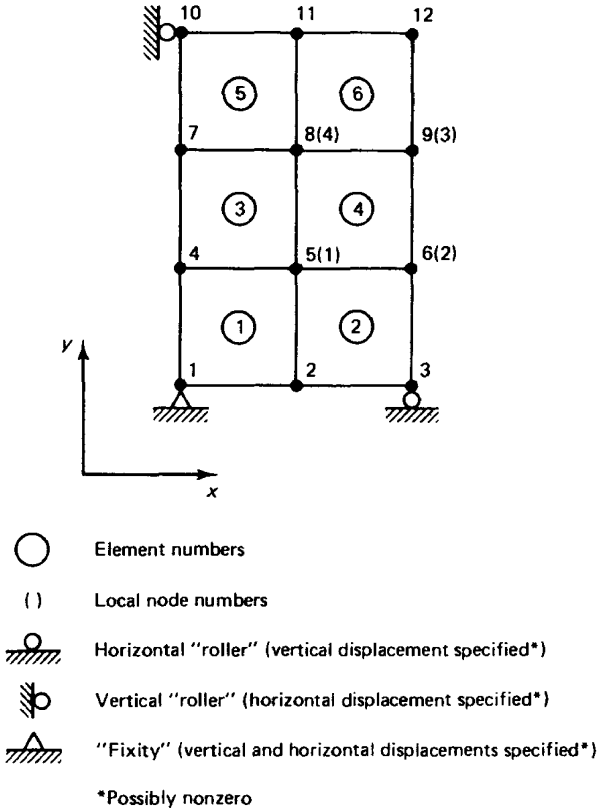


Figure 2.10.1 Mesh of four-node, rectangular, elasticity elements; global and local node numbers, element numbers, and displacement boundary conditions.

In terms of the IEN and LM arrays, a precise definition of the q_p^e 's may be given (see (2.9.5)):

$$q_p^e = q_{ia}^e = \begin{cases} 0, & \text{if } LM(i, a, e) \neq 0 \\ q_{iA}, & \text{where } A = IEN(a, e), \text{ if } LM(i, a, e) = 0 \end{cases} \quad (2.10.1)$$

This definition may be easily programmed.

¹⁸ In practice, equation numbers are often renumbered internally to minimize the bandwidth of and thus decrease storage and solution effort. This is especially important in analyzing large-scale systems involving tens of thousands of equations. An algorithm for reducing bandwidth is presented in [8].

ID array:

		Global node numbers (A)												$(n_{np} = 12)$
		1	2	3	4	5	6	7	8	9	10	11	12	
Global degrees of freedom (i)	1	0*	1	3	4	6	8	10	12	14	0	17	19	$(n_{eq} = 20)$
	2	0	2	0	5	7	9	11	13	15	16	18	20	

$P = ID(i, A)$

$(n_{dof} = 2)$

IEN array:

		Element numbers (e)						$(n_{el} = 6)$
		1	2	3	4	5	6	
Local node numbers (a)	1	1	2	4	5	7	8	$(n_{en} = 4)$
	2	2	3	5	6	8	9	
	3	5	6	8	9	11	12	
	4	4	5	7	8	10	11	

$A = IEN(a, e)$

LM array:

		Element numbers (e)						$(n_{el} = 6)$
		1	2	3	4	5	6	
1	1	0*	1	4	6	10	12	1
	2	0	2	5	7	11	13	
	3	1	3	6	8	12	14	2
	4	2	0	7	9	13	15	
	5	6	8	12	14	17	19	3
	6	7	9	13	15	18	20	
	7	4	6	10	12	0	17	4
	8	5	7	11	13	16	18	

$P = LM(p, e) = LM(i, a, e) = ID(i, IEN(a, e))$

$(n_{ee} = 8)$

Element degree of freedom numbers (i, where $1 \leq i \leq n_{ed}$ and here $n_{ed} = 2$)

Local node numbers (a)

Local equation numbers ($p = n_{ed}(a - 1) + i$)

*Displacement boundary conditions are denoted by zeros.

Figure 2.10.2 ID, IEN, and LM arrays for the mesh of Figure 2.10.1.

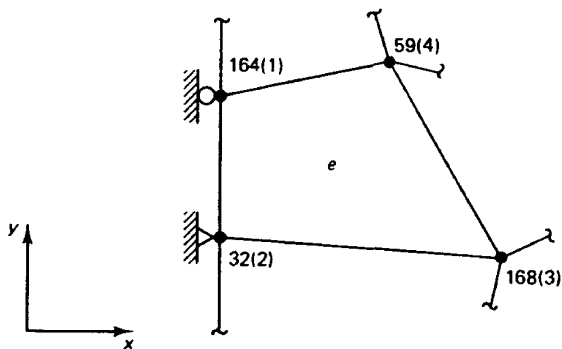
Example 2

As a final example, we consider a typical four-node, elasticity element in some large mesh; see Fig. 2.10.3. We assume the pertinent entries of the ID array are given follows:

$$\left. \begin{aligned} \text{ID}(1, 32) &= 0 \\ \text{ID}(2, 32) &= 0 \\ \text{ID}(1, 59) &= 115 \\ \text{ID}(2, 59) &= 116 \\ \text{ID}(1, 164) &= 0 \\ \text{ID}(2, 164) &= 325 \\ \text{ID}(1, 168) &= 332 \\ \text{ID}(2, 168) &= 333 \end{aligned} \right\} \quad (2.10.3)$$

The entries of IEN follow from Fig. 2.10.3:

$$\left. \begin{aligned} \text{IEN}(1, e) &= 164 \\ \text{IEN}(2, e) &= 32 \\ \text{IEN}(3, e) &= 168 \\ \text{IEN}(4, e) &= 59 \end{aligned} \right\} \quad (2.10.4)$$



() — Local node numbers

Figure 2.10.3 Typical four-node elasticity element; global and local node numbers.

Combining (2.10.4) and (2.10.5), by way of (2.10.1), yields entries of the LM array:

$$\left. \begin{aligned} \text{LM}(1, 1, e) &= 0 \\ \text{LM}(2, 1, e) &= 325 \\ \text{LM}(1, 2, e) &= 0 \\ \text{LM}(2, 2, e) &= 0 \\ \text{LM}(1, 3, e) &= 332 \\ \text{LM}(2, 3, e) &= 333 \\ \text{LM}(1, 4, e) &= 115 \\ \text{LM}(2, 4, e) &= 116 \end{aligned} \right\} \quad (2.10.6)$$

The contribution to the global arrays may be deduced from LM:

Stiffness (due to symmetry, only the upper triangular portion need be assembled.)

$$\left. \begin{aligned} K_{115, 115} &\leftarrow K_{115, 115} + k_{77}^e \\ K_{115, 116} &\leftarrow K_{115, 116} + k_{78}^e \\ K_{115, 325} &\leftarrow K_{115, 325} + k_{72}^e \\ K_{115, 332} &\leftarrow K_{115, 332} + k_{75}^e \\ K_{115, 333} &\leftarrow K_{115, 333} + k_{76}^e \\ \\ K_{116, 116} &\leftarrow K_{116, 116} + k_{88}^e \\ K_{116, 325} &\leftarrow K_{116, 325} + k_{82}^e \\ K_{116, 332} &\leftarrow K_{116, 332} + k_{85}^e \\ K_{116, 333} &\leftarrow K_{116, 333} + k_{86}^e \\ \\ K_{325, 325} &\leftarrow K_{325, 325} + k_{22}^e \\ K_{325, 332} &\leftarrow K_{325, 332} + k_{25}^e \\ K_{325, 333} &\leftarrow K_{325, 333} + k_{26}^e \\ \\ K_{332, 332} &\leftarrow K_{332, 332} + k_{55}^e \\ K_{332, 333} &\leftarrow K_{332, 333} + k_{56}^e \\ \\ K_{333, 333} &\leftarrow K_{333, 333} + k_{66}^e \end{aligned} \right\} \quad (2.10.7)$$

Force

$$\left. \begin{aligned} F_{115} &\leftarrow F_{115} + f_7^e \\ F_{116} &\leftarrow F_{116} + f_8^e \\ F_{325} &\leftarrow F_{325} + f_2^e \\ F_{332} &\leftarrow F_{332} + f_5^e \\ F_{333} &\leftarrow F_{333} + f_6^e \end{aligned} \right\} \quad (2.10.8)$$

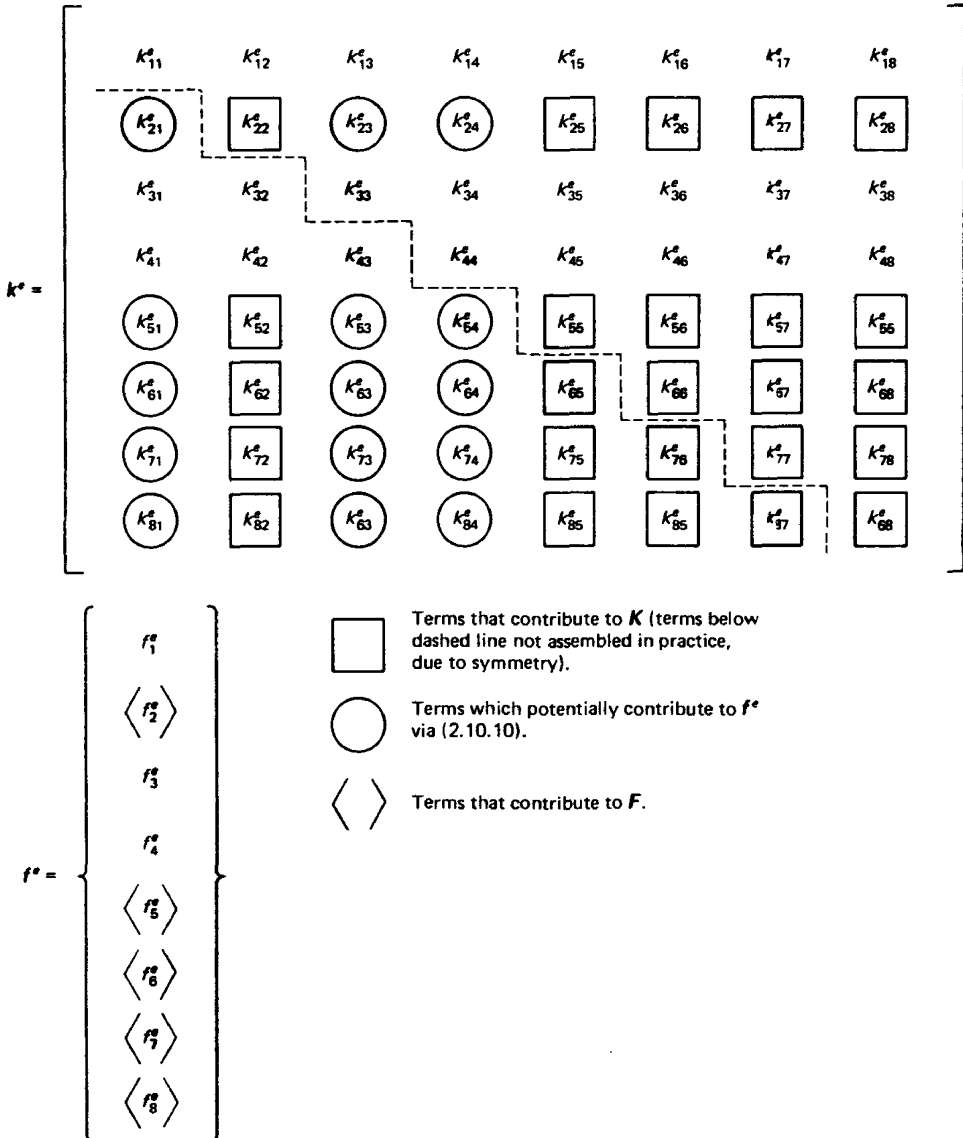
where

$$f_p^e = \cdots - \sum_{q=1}^{n_{ec}} k_{pq}^e q_q^e \quad (2.10.9)$$

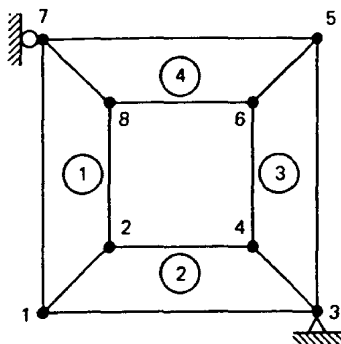
(We have omitted the first two terms in the right-hand side of (2.9.5) in writing (2.10.9).) In the present example, only q_1^e , q_3^e and q_4^e may be nonzero. Therefore (2.10.9) may be simplified to

$$f_p^e = \cdots - k_{p1}^e q_1^e - k_{p3}^e q_3^e - k_{p4}^e q_4^e \quad (2.10.10)$$

The multiplications indicated in (2.10.10) are only performed in practice if the q_p^e 's are nonzero. A schematic representation of the contributions of k^e and f^e to K and F is shown in Figure 2.10.4.



Exercise 1. Consider a two-dimensional elastostatic boundary-value problem. Set up the ID, IEN, and LM arrays for the following mesh of four-node quadrilaterals:



2.11 SUMMARY OF IMPORTANT EQUATIONS FOR PROBLEMS CONSIDERED IN CHAPTERS 1 AND 2

1. Classical Linear Elastostatics

$$\begin{aligned}
 (S) \quad & \sigma_{ij,j} + f_i = 0 && \text{on } \Omega \\
 & u_i = q_i && \text{on } \Gamma_{q_i} \\
 & \sigma_{ij}n_j = h_i && \text{on } \Gamma_{h_i}
 \end{aligned}$$

where $\sigma_{ij} = c_{ijkl}\epsilon_{kl} = c_{ijkl}u_{(k,l)}$

(W)¹⁹ Find $u \in \mathcal{S}$, such that $\forall w \in \mathcal{O}$

$$a(w, u) = (w, \ell) + (w, h)_\Gamma$$

where

$$a(w, u) = \int_{\Omega} w_{(i,j)} c_{ijkl} u_{(k,l)} d\Omega$$

$$(w, \ell) = \int_{\Omega} w_i \ell_i d\Omega$$

$$(w, h)_\Gamma = \sum_{i=1}^{n_{\text{ed}}} \left(\int_{\Gamma_{h_i}} w_i h_i d\Omega \right)$$

(G) Find $v^h \in \mathcal{O}^h$, such that $\forall w^h \in \mathcal{O}^h$

$$a(w^h, v^h) = (w^h, \ell) + (w^h, h)_\Gamma - a(w^h, q^h)$$

¹⁹The notation $\forall$ means "for all."

$$(M) \quad Kd = F, \text{ where } K = \mathbf{A}_{e=1}^{n_{el}}(k^e), F = F_{\text{nodal}} + \mathbf{A}_{e=1}^{n_{el}}(f^e)^{20}$$

$$k_{pq}^e = e_i^T k_{ab}^e e_j, \quad k_{ab}^e = \int_{\Omega^e} B_a^T D B_b d\Omega$$

$$f_p^e = \int_{\Omega^e} N_a \ell_i d\Omega + \int_{\Gamma_k} N_a h_i d\Gamma - \sum_{q=1}^{n_{el}} k_{pq}^e \varphi_q^e \quad (\text{no sum on } i)$$

$$p = n_{ed}(a - 1) + i$$

$$q = n_{ed}(b - 1) + j$$

$$\text{Stress at a point: } \sigma(x) = D(x) \sum_{a=1}^{n_{en}} B_a(x) d_a^e$$

2. Classical Linear Heat Conduction

$$(S) \quad \begin{aligned} q_{i,i} &= \ell & \text{in } \Omega \\ u &= q & \text{on } \Gamma_q \\ -q_i n_i &= h & \text{on } \Gamma_h \end{aligned}$$

$$\text{where } q_i = \kappa_{ij} u_{,j}$$

$$(W) \quad \text{Find } u \in \mathcal{S}, \text{ such that } \forall w \in \mathcal{U}$$

$$a(w, u) = (w, \ell) + (w, h)_\Gamma$$

where

$$a(w, u) = \int_{\Omega} w_{,i} \kappa_{ij} u_{,j} d\Omega$$

$$(w, \ell) = \int_{\Omega} w \ell d\Omega$$

$$(w, h)_\Gamma = \int_{\Gamma} w h d\Gamma$$

$$(G) \quad \text{Find } v^h \in \mathcal{S}^h, \text{ such that } \forall w^h \in \mathcal{U}^h$$

$$a(w^h, v^h) = (w^h, \ell) + (w^h, h)_\Gamma - a(w^h, q^h)$$

²⁰In defining F we have added to the element contributions the term F_{nodal} , which is a vector of nodal applied forces. The reason for this is that it is often easier in practice to directly input concentrated forces at nodes rather than go through the element-by-element form and assemble procedure. The expression for F then emphasizes that both modes of constructing F are to be accommodated in the computer implementation of problems of this type.

$$(M) \quad Kd = F, \text{ where } K = \mathbf{A}_{e=1}^{n_{el}}(k^e), \quad F = F_{\text{nodal}} + \mathbf{A}_{e=1}^{n_{el}}(f^e)^{20}$$

$$k_{ab}^e = \int_{\Omega^e} B_a^T D B_b \, d\Omega$$

$$f_a^e = \int_{\Omega^e} N_a \ell \, d\Omega + \int_{\Gamma_k^e} N_a h \, d\Gamma - \sum_{b=1}^{n_{el}} k_{ab}^e q_b^e$$

$$\text{Heat flux vector at a point: } q(x) = -D(x) \sum_{a=1}^{n_{en}} B_a(x) d_a^e$$

3. One-Dimensional Model Problem

$$(S) \quad \begin{aligned} u_{,xx} + \ell &= 0 & \text{on } \Omega &=]0, 1[\\ u(1) &= q & (\Gamma_q &= \{1\}) \\ -u_{,x}(0) &= h & (\Gamma_h &= \{0\}) \end{aligned}$$

$$(W) \quad \text{Find } u \in \mathcal{S}, \text{ such that } \forall w \in \mathcal{U}$$

$$a(w, u) = (w, \ell) + wh(0)$$

where

$$a(w, u) = \int_0^1 w_{,x} u_{,x} \, dx$$

$$(w, \ell) = \int_0^1 w \ell \, dx$$

$$(G) \quad \text{Find } v^h \in \mathcal{U}^h, \text{ such that } \forall w^h \in \mathcal{U}^h$$

$$a(w^h, v^h) = (w^h, \ell) + w^h(0)h - a(w^h, q^h)$$

$$(M) \quad Kd = F, \text{ where } K = \mathbf{A}_{e=1}^{n_{el}}(k^e), \quad F = F_{\text{nodal}} + \mathbf{A}_{e=1}^{n_{el}}(f^e)^{20}$$

$$k_{ab}^e = \int_{\Omega^e} N_{a,x} N_{b,x} \, dx$$

$$f_a^e = \int_{\Omega^e} N_a \ell \, dx + \begin{cases} h \delta_{a1} & e = 1 \\ 0 & e = 2, \dots, n_{el} - 1 \\ -q k_{2a}^e & e = n_{el} \end{cases}$$

2.12 AXISYMMETRIC FORMULATIONS AND ADDITIONAL EXERCISES

Axisymmetric formulations are expressed in terms of cylindrical coordinates:

$$x_1 = r = \text{the radial coordinate}$$

$$x_2 = z = \text{the axial coordinate}$$

$$x_3 = \theta = \text{the circumferential coordinate}$$

The basic hypothesis of axisymmetry is that all functions under consideration are independent of θ . That is, they are functions of r and z only. Thus three-dimensional problem classes are reduced to two-dimensional ones.

Heat Conduction

The axisymmetric formulation for heat conduction is almost identical to the two-dimensional Cartesian case considered previously. The only difference is that a factor of $2\pi r$ needs to be included in each integrand of the variational equation to account for the correct volumetric weighting (e.g., $d\Omega = 2\pi r dr dz$ replaces $d\Omega = dx_1 dx_2$). Since the constant 2π is common to all terms, it may be cancelled throughout if desired.

Elasticity

The displacement components in cylindrical coordinates are:

$$u_1 = u_r = \text{the radial displacement}$$

$$u_2 = u_z = \text{the axial displacement}$$

$$u_3 = u_\theta = \text{the circumferential displacement}$$

In addition to the basic hypothesis of axisymmetry, we further assume that $u_\theta = 0$, and thus

$$\epsilon_{r\theta} = \epsilon_{z\theta} = 0 \quad (2.12.1)$$

Note that $\epsilon_{\theta\theta} = u_r/r$ and therefore it is generally not zero. The constitutive moduli are assumed to be such that the preceding kinematical hypotheses result in

$$\sigma_{r\theta} = \sigma_{z\theta} = 0 \quad (2.12.2)$$

The preceding assumptions lead to what is called the *torsionless axisymmetric case*. This formulation is similar to but somewhat more complicated than the two-dimensional cases of plane strain and plane stress. Here there are four nonzero components of stress and strain:

$$\sigma = \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \\ \sigma_{33} \end{Bmatrix} = \begin{Bmatrix} \sigma_{rr} \\ \sigma_{zz} \\ \sigma_{rz} \\ \sigma_{\theta\theta} \end{Bmatrix} \quad (2.12.3)$$

$$\epsilon = \begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ 2\epsilon_{12} \\ \epsilon_{33} \end{Bmatrix} = \begin{Bmatrix} \epsilon_{rr} \\ \epsilon_{zz} \\ 2\epsilon_{rz} \\ \epsilon_{\theta\theta} \end{Bmatrix} \quad (2.12.4)$$

The ordering emanates from the following generalization of Table 2.7.1:

$I \backslash J$	$i \backslash k$	$j \backslash l$
1	1	1
3	1	2
3	2	1
2	2	2
4	3	3

The D array takes on the following form:

$$D = [D_{IJ}] = \underbrace{\begin{bmatrix} D_{33} & D_3 \\ D_3^T & D_{44} \end{bmatrix}}_{4 \times 4} \quad (2.12.5)$$

$$D_{33} = \begin{bmatrix} D_{11} & D_{12} & D_{13} \\ & D_{22} & D_{23} \\ \text{symmetric} & & D_{33} \end{bmatrix} \quad (2.12.6)$$

$$D_3 = \begin{Bmatrix} D_{14} \\ D_{24} \\ D_{34} \end{Bmatrix} \quad (2.12.7)$$

where $D_{IJ} = c_{ijkl}$, in which the indices are related by the table. The B_a -matrix takes on the form

$$B_a = \left[\begin{array}{cc|c} N_{a,1} & 0 & \\ 0 & N_{a,2} & \\ N_{a,2} & N_{a,1} & \\ \hline \frac{N_a}{r} & 0 & \end{array} \right] \quad (2.12.8)$$

Again, a factor of $2\pi r$ needs to be included in all integrands.

Remark

The *plane strain* case may be obtained from the axisymmetric formulation by

- i. Ignoring the $2\pi r$ factors; and
- ii. Ignoring the fourth row of B_a and the fourth row and column of D .

Furthermore, the *plane stress* case may be similarly obtained if, in addition, D_{33} is replaced by

$$D_{33} - D_3 D_{44}^{-1} D_3^T \quad (2.12.9)$$

which directly follows from the plane stress condition, $\sigma_{44} = 0$. (References [6] and [7] may be consulted for further elaboration on the physical ideas.) Sometimes (2.12.9) is referred to as the *statically condensed* elastic coefficient matrix.

Consequently, in programming the axisymmetric case, for a small amount of additional effort both plane strain and plane stress may also be included.

Exercise 1. Under the assumption of isotropy, show that D_{33} is the same as the D -matrix in (2.7.34). Furthermore, show that $D_{44} = \lambda + 2\mu$ and

$$D_3 = \begin{Bmatrix} \lambda \\ \lambda \\ 0 \end{Bmatrix} \quad (2.12.10)$$

Exercise 2. Verify that for the isotropic case, (2.12.9) achieves a similar end to the procedure described in Remark 9 of Sec. 2.7.

Exercise 3. Consider the one-dimensional model problem discussed previously. Obtain exact expressions for $f = \{f_a^e\}$, $a = 1, 2$, for the following cases (ignore q and h contributions):

- i. $\ell = \text{constant}$.
- ii. $\ell = \delta(x - \bar{x})$, the delta function, where $x_1^e \leq \bar{x} \leq x_2^e$. Specialize for the cases $\bar{x} = x_b^e$ and $\bar{x} = (x_1^e + x_2^e)/2$.

Solution

$$f_a^e = \int_{x_1^e}^{x_2^e} N_a(x) \ell(x) dx$$

$$\text{i. } f_a^e = \ell \int_{x_1^e}^{x_2^e} N_a(x) dx = \frac{\ell h^e}{2} \underbrace{\int_{-1}^{+1} N_a(\xi) d\xi}_1$$

$$f^e = \frac{\ell h^e}{2} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$$

$$\text{ii. } f_a^e = \int_{x_1^e}^{x_2^e} N_a(x) \delta(x - \bar{x}) dx = N_a(\bar{x})$$

For $x = x_b^e$,

$$f_a^e = N_a(\bar{x}) = N_a(x_b^e) = \delta_{ab} \quad (\text{Kronecker delta})$$

$$f^e = \begin{Bmatrix} \delta_{1b} \\ \delta_{2b} \end{Bmatrix}$$

For $\bar{x} = (x_1^e + x_2^e)/2$,

$$f_a^e = N_a(\bar{x}) = N_a\left(\frac{x_1^e + x_2^e}{2}\right) = \frac{1}{2}$$

Therefore,

$$f^e = \frac{1}{2} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$$

Exercise 4. Consider the boundary-value problem for classical linear elastostatics discussed previously. In the linearized theory of *small displacements superposed upon large*, the stiffness term in the variational equation,

$$\int_{\Omega} w_{(i,j)} c_{ijkl} u_{(k,l)} d\Omega$$

is replaced by

$$\int_{\Omega} w_{i,j} d_{ijkl} u_{k,l} d\Omega$$

where

$$d_{ijkl} = c_{ijkl} + \delta_{ik} \sigma_{jl}^0$$

$$\sigma_{jl}^0 = \sigma_{lj}^0$$

and the σ_{jl}^0 's (i.e., *initial stresses*) are given functions of $\mathbf{x} \in \Omega$. It follows from the symmetries of c_{ijkl} and σ_{jl}^0 that

$$d_{ijkl} = d_{klij}$$

Assume $n_{sd} = 2$. An index-free formulation of the stiffness term is given by

$$\int_{\Omega} \begin{Bmatrix} w_{1,1} \\ w_{2,2} \\ w_{1,2} + w_{2,1} \\ w_{1,2} - w_{2,1} \end{Bmatrix}^T \underbrace{\mathbf{D}}_{4 \times 4} \begin{Bmatrix} u_{1,1} \\ u_{2,2} \\ u_{1,2} + u_{2,1} \\ u_{1,2} - u_{2,1} \end{Bmatrix} d\Omega$$

which leads to the following definition of the element stiffness matrix:

$$k_{pq}^e = \mathbf{e}_i^T \int_{\Omega^e} \underbrace{\mathbf{B}_a^T}_{2 \times 4} \underbrace{\mathbf{D}}_{4 \times 4} \underbrace{\mathbf{B}_b}_{4 \times 2} d\Omega \mathbf{e}_j$$

Set up $\mathbf{B}_a$ in terms of the shape function N_a . Define the components of $\mathbf{D}$ in terms of the d_{ijkl} 's. (The σ_{jl}^0 -contribution to the stiffness is sometimes called the *initial-stress stiffness matrix*. It is important to account for it in the solution of many nonlinear problems.)

Exercise 5. Let Ω be a region in $\mathbb{R}^2$ and let its boundary $\Gamma = \overline{\Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_4}$ where $\Gamma_1, \dots, \Gamma_4$ are nonoverlapping subregions of Γ . Let $\mathbf{n}$ be the unit outward normal vector to Γ such that $\mathbf{s}$ and $\mathbf{n}$ form a right-hand rule basis; see Fig. 2.12.1. Consider the following boundary-value problem in classical linear elastostatics: Given $\ell_i : \Omega \rightarrow \mathbb{R}$; $q_i : \Gamma_1 \rightarrow \mathbb{R}$; $h_i : \Gamma_2 \rightarrow \mathbb{R}$; q_n and $h_s : \Gamma_3 \rightarrow \mathbb{R}$; and q_s and $h_n : \Gamma_4 \rightarrow \mathbb{R}$; find $u_i : \Omega \rightarrow \mathbb{R}$ such that

$$\begin{aligned} \sigma_{ij,j} + f_i &= 0 && \text{in } \Omega \\ u_i &= q_i && \text{on } \Gamma_1 \\ \sigma_{ij}n_j &= h_i && \text{on } \Gamma_2 \\ \left. \begin{aligned} u_i n_i &= q_n \\ \sigma_{ij}n_j s_i &= h_s \end{aligned} \right\} && \text{on } \Gamma_3 && \begin{pmatrix} \text{normal displacement} \\ \text{tangential traction} \end{pmatrix} \\ \left. \begin{aligned} u_i s_i &= q_s \\ \sigma_{ij}n_j n_i &= h_n \end{aligned} \right\} && \text{on } \Gamma_4 && \begin{pmatrix} \text{tangential displacement} \\ \text{normal traction} \end{pmatrix} \end{aligned}$$

where $\sigma_{ij} = c_{ijkl}u_{(k,l)}$. Establish a weak formulation for this problem in which all “ q -type” boundary conditions are essential and all “ h -type” boundary conditions are natural. State all requirements on the spaces $\mathcal{S}$ and $\mathcal{O}$. *Hint:* $\mathbf{w} = w_n \mathbf{n} + w_s \mathbf{s}$; i.e., $w_i = w_n n_i + w_s s_i$.

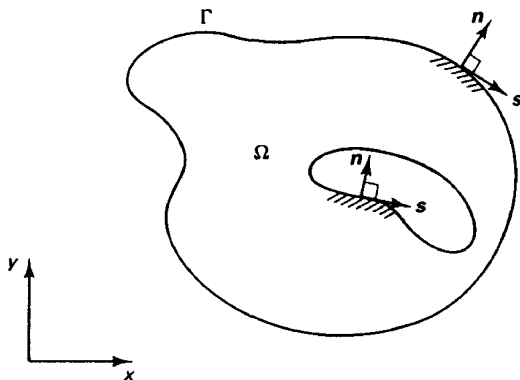


Figure 2.12.1

Exercise 6. In practice, it is often useful to generalize the constitutive equation of classical elasticity to the form

$$\sigma_{ij} = c_{ijkl}(\epsilon_{kl} - \epsilon_{kl}^0) + \sigma_{ij}^0 \quad (2.12.11)$$

where ϵ_{ij}^0 and σ_{ij}^0 are the *initial strain* and *initial stress*, both given functions of $\mathbf{x}$. The initial strain term may be used to represent thermal expansion effects by way of

$$\epsilon_{kl}^0 = -\theta c_{kl} \quad (2.12.12)$$

where θ is the *temperature* and the c_{kl} 's are the *thermal expansion coefficients* (both given functions). Clearly, (2.12.11) will in no way change the stiffness matrix. However there will be additional contributions to f_p^e . Generalize the definition of f_p^e to account for these additional terms.

Solution We begin with the weak form

$$\int_{\Omega} w_{(i,j)} \sigma_{ij} d\Omega = \int_{\Omega} w_i \ell_i d\Omega + \sum_{i=1}^{n_{sd}} \left(\int_{\Gamma_{h_i}} w_i h_i d\Gamma \right)$$

Substituting (2.12.11) and (2.12.12) into the weak form leads to

$$\int_{\Omega} w_{(i,j)} c_{ijkl} \epsilon_{kl} d\Omega = \int_{\Omega} w_i \ell_i d\Omega + \sum_{i=1}^{n_{sd}} \left(\int_{\Gamma_{k_i}} w_i h_i d\Gamma \right)$$

$\left\{ \begin{array}{l} \text{As before} \\ \text{New contributions} \\ \text{to right-hand side:} \end{array} \right.$

$$+ \int_{\Omega} w_{(i,j)} c_{ijkl} \epsilon_{kl}^0 d\Omega - \int_{\Omega} w_{(i,j)} \sigma_{ij}^0 d\Omega$$

from which the additional terms in f_p^e may be deduced:

$$f_p^e = \dots + e_i^T \int_{\Omega^e} B_a^T D \theta c d\Omega - e_i^T \int_{\Omega^e} B_a^T \sigma^0 d\Omega$$

$\uparrow$
 Anywhere
 inside integral

where

$$c = \begin{Bmatrix} c_{11} \\ c_{22} \\ 2c_{12} \end{Bmatrix}; \quad \sigma^0 = \begin{Bmatrix} \sigma_{11}^0 \\ \sigma_{22}^0 \\ \sigma_{12}^0 \end{Bmatrix}; \dots$$

$\uparrow$
 Without loss of generality, we
 may assume symmetry, i.e., $c_{12} + c_{21} = 2c_{12}$

Exercise 7. Consider the following boundary-value problem:

$$u_{,xx} - p(p-1)x^{p-2} = 0, \quad 0 < x < 1$$

$$-u_{,x}(0) = 0$$

$$u(1) = 1$$

where p is a given constant.

- i. Obtain the exact solution to this problem for $p = 5$. Sketch.
- ii. State the weak formulation of the problem.
- iii. State the Galerkin formulation.
- iv. State the matrix formulation.
- v. Solve the matrix problem assuming $p = 5$ and using the piecewise linear finite element space for the following cases:
 - a. one element
 - b. two equal-length elements
- vi. Compare the exact value of $u_{,x}(1)$ with the approximate values computed in part v. Explain why it is impossible for these results to compare favorably.

Exercise 8. In heat conduction, it is often of interest to accurately calculate the boundary heat flux over a portion of the boundary where temperature is specified. Suppose we use the usual Galerkin finite element formulation to calculate the temperature. However, instead of calculating the heat flux in the usual way (i.e., by differentiating the temperature), we introduce a *post-processing* which derives from the following weak formulation:

Find $u \in \mathcal{S}$ and $h \in L_2(\Gamma_\theta)$ such that for all $w \in \mathcal{U}$,

$$-\int_{\Omega} w_{,i} q_i d\Omega = \int_{\Omega} w f d\Omega + \int_{\Gamma_A} w h d\Gamma + \int_{\Gamma_\theta} w h d\Gamma$$

where h is the *unknown* heat flux on Γ_θ

(Note: In this formulation, it is *not* assumed that $w = 0$ on Γ_θ !)

- i. Show, in addition to the usual differential equations and boundary conditions, that

$$h = -q_i n_i \quad \text{on } \Gamma_\theta$$

arises naturally from the new weak formulation.

- ii. State the Galerkin and matrix formulations corresponding to the new weak formulation assuming h is approximated in the usual way, namely

$$h^h(x) = \sum_{A \in \eta_\theta} N_A(x) h_A$$

(Hint: The equations governing the temperature are unchanged.)

- iii. Specialize this formulation to the one-dimensional problem described in Exercise 7 and calculate the boundary flux at $x = 1$ by the new procedure (cf. parts v and vi of Exercise 7).

(Hint: The new method should produce exact results for these cases.)

- iv. Develop a counterpart of the new formulation for elasticity. That is, introduce the i th component of traction as an independent unknown on Γ_θ , and carefully state the weak formulation.
- v. Prove that the new method is exact for the one-dimensional model problem of Chapter 1.

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